

Hadamard PTQ README

Goal

This is an experimental setup to observe how **Hadamard rotation** affects **PTQ (Post-Training Quantization)**.

Here, `alpha` should be treated as an **experimental knob** rather than a theoretically optimal parameter. It is mainly used to explore how scaling/rotation influences quantization error.

How to Run

Just run:

- `software/hadamard_PTQ`

This script is sufficient to feel/observe the Hadamard effect (e.g., error metrics, accuracy changes, MSE comparisons—depending on what the script prints).

Why Hadamard Can Help (Intuition)

Hadamard rotation helps PTQ mainly by:

- applying an **orthogonal “rotation”** (typically along the channel dimension),
- making the activation/weight distribution **less dominated by outliers**,
- reducing how much outliers determine the quantization scale and clipping,
- therefore potentially **reducing PTQ error**.

In short: **rotation spreads out outliers, which can make quantization easier**.

Important : Unstable Gains on Small Networks (e.g., ResNet)

On relatively small networks such as **ResNet**, the improvement can be **unstable**:

- sometimes it improves,
- sometimes it changes little,
- and **sometimes it can be worse than without rotation**.

This happens because small networks can be more sensitive to distribution shifts and error propagation, and outliers are not always the main bottleneck for PTQ.